Workshop Pole Barn: Digging the Footings

Mini excavator holes and the forms that go in them · 19 building piers · Detail 3 · Prairie, Idaho · planned for 2028

Dig and form to Detail 3. Do not shrink the hole to fit the auger. Each building pier is an 18 in tube on a 30 in round pad, 12 in thick, with the bottom 60 in below the pad surface. An 18 in auger that stops at 4 ft cannot make that pier. A mini excavator can. The engineer still has to accept the pier, the bars in it, and any hole that is not this size. This page is the digging and forming sequence, not a stamped design.

The hole

Measure depth down from the finished pad surface, not from the pile of spoil and not from the day the fill was placed. The north edge of the pad is almost 2 ft of imported fill and the south edge is 6 in to 1 ft, so a change in soil color is not the bottom. Keep digging until the bottom is 60 in below the pad.

One building pier, from Detail 3

Part	Size	Where it sits
Hole, dug	60 in deep, about 3 ft across at the bottom	Centered on the post mark. Wider if the banks will not stand. The extra width is backfill, not concrete.
Pad form	30 in outside diameter, 12 in tall	On the cleaned bottom. This ring, not the dirt wall, is the concrete.
Tube	18 in round, about 48 in long, open at the bottom	Open end at the top of the pad, braced plumb. It does not go down into the pad. Its top is at the pad surface.
Concrete	4,000 psi, about 0.44 cu yd each	One pour fills the ring and the tube. About 9 cu yd for the 19 building piers, with a little waste.
Bracket	Wet-set, sized for a 3-ply 2×8	Set in the wet top. The concrete stops at the pad surface. The post, built on the post schedule, stands on the bracket and does not touch concrete.

A 3 ft bottom gives a few inches of room around the 30 in ring so it can be centered and plumbed from above. Do not dig an 18 in hole and hope the pad can be added later. A 30 in pad will not go down an 18 in hole, and a pad poured at the top of the hole does not spread the load into the soil at the bottom.

Figure 1. The 18 in tube stops at the top of the pad. It does not run down through it. The pad is a solid 30 in disk. The tube is a paper sleeve with no bottom, so the column concrete lands on that disk.

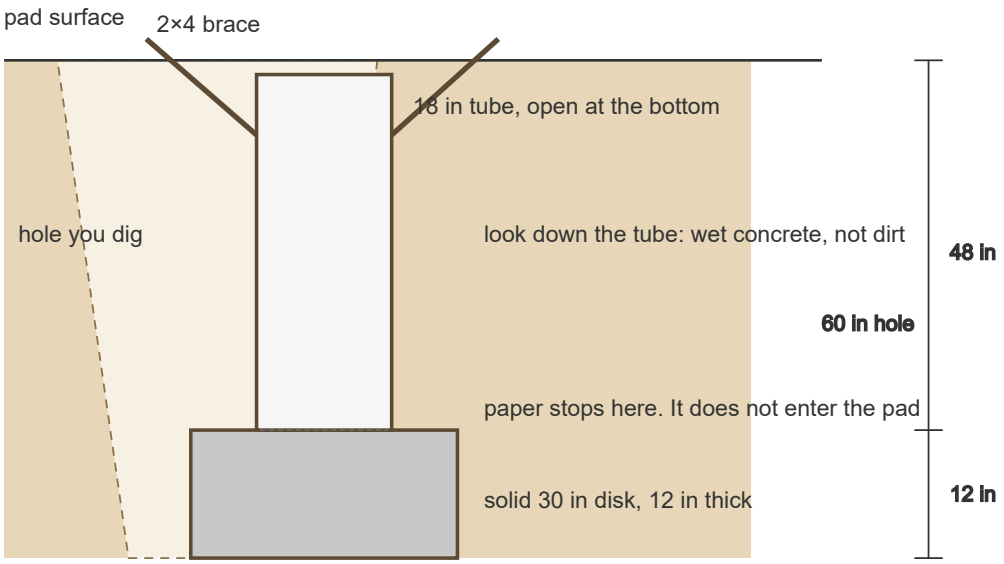
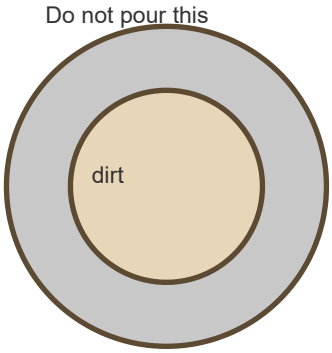
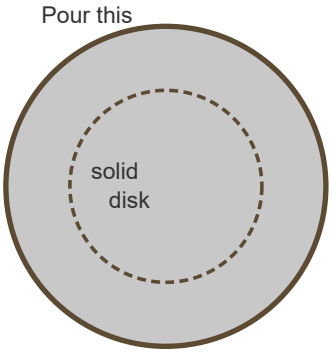


Figure 2. Looking down. The left pour is a doughnut: paper on the dirt, concrete only in the ring, column sitting on dirt. The right pour is the pad: a solid disk, then the open tube on top of it.



paper tube on the dirt
concrete only in the ring



30 in pad, no hole in it
dashed circle is the open tube on top

A doughnut is what you get if the 18 in tube stands on the dirt inside the 30 in ring. The truck then fills only the space between the two paper walls. The column sits on dirt. The paper keeps that ring of concrete off the column. That is not a 30 in pad, and it is not this pier.

A sonotube has no bottom. It is a paper sleeve. Brace it so the open end is at the top of the ring, not down in the ring. Fill the ring from beside the tube until you look down the tube and see wet concrete, not dirt. That concrete is a solid 30 in disk, and the paper never passed through it. Then fill the tube. The column concrete lands on that wet disk across the whole open 18 in bottom. Do not push the paper down into the pad. Pushing it in is what puts cardboard in the joint.

Before the first hole

- **Locate pipes.** Call 811 before digging. The well branch runs east of this pad. Know where that line, and any power, actually is. A 5 ft hole is deep enough to find a pipe the hard way.
- **String the post lines first.** Set batter boards outside the digging, where the excavator will not knock them over. Every hole is centered on a post mark from the plans. An excavator hole is wider than the pier, so the tube, not the hole, has to land on that mark.
- **Have the forms and the brackets on site** before the truck is ordered. A hole left open for a week can cave in or fill with water, and a truck waiting on forms is a wasted load.

Dig, then set the form

1. Dig with the mini excavator to 60 in below the pad surface. Keep the bottom flat. Scrape loose crumbs off it so the pad bears on firm native soil, not on fill that fell back in. Pile the spoil back from the edge so it cannot roll in or load the bank.
2. Do not climb into a straight-sided 5 ft hole. Set the form from above. If someone has to go in, slope the banks back first so the wall is not vertical. This hole is deep enough to bury a person if a bank lets go.
3. Build the pad form before dig day: a round ring, 30 in outside and 12 in tall. A 12 in slice of 30 in form tube does it. A square box is not this pier. The drawing is a 30 in round pad.
4. Set the 30 in ring on the cleaned bottom, centered on the post mark. Leave the middle empty. This ring is the outside form. It is not the shape of the hole, and the 18 in tube does not stand inside it on the dirt.
5. Brace the 18 in tube so its open bottom is at the top of that ring, centered and plumb on the string. The paper stops there. It does not go down into the ring. Cut the tube about 48 in so the top lands on the pad-surface string. One 12 ft tube still makes two. Cut off any cap or taped end. A 2×4 across the top of the hole, and a second brace sideways, hold the tube. The wet concrete will not hold it. Do not let go until the braces are on.
6. Plumb the tube and brace it in two directions, from stakes outside the hole. Check it against the string. Concrete will shove an unbraced tube off the post line.
7. Pump out standing water. Do not pour into a hole with water in the bottom.

The pour, then the backfill

1. Order 4,000 psi for the pier pour, separate from the later slab pour, as the materials list already has it. About 9 cu yd covers the 19 building piers if every form is ready. The chute will not reach the middle of a 32 by 48 ft building, so plan a pump, wheelbarrows, or a path the truck can use.
2. Fill the ring first, from beside the tube, in the same truck visit as the tube. Keep pouring until you look down the tube and see wet concrete, not dirt. Rod the ring if the mix sits in the gap and will not show inside the tube. Then fill the tube. Do not pour only in the gap between the two paper walls. Do not backfill an empty tube and then call the truck. Dirt around an empty paper tube crushes it.
3. Strike the concrete off at the pad surface. Set the wet-set bracket in that wet top, on the string, at the embed the bracket maker specifies. Have all 19 brackets at the hole. The concrete will not wait.
4. Tie the bars in Figures 3 and 4 and hang them in the tube before the truck arrives, with the hooks down in the ring and 3 in off the soil. The tube stays at the top of the ring. The bars hang below it. Keep them off the rebar legs of the bracket. If a leg hits a bar, slide the bar. Do not cut a leg,

and do not pour until the bars are in the form.

5. The next day, after the concrete has set, backfill the space outside the tube and the ring. Place it in layers and tamp each layer. Loose backfill settles under the slab edge. The slab comes later, with an isolation strip against the pier, as Detail 3 draws it.

The bracket, and why the post stays dry

The wood post does not go in the concrete. The piece that goes in the wet pier is a steel bracket. Its rebar legs sink into the fresh concrete. Its U-shaped pocket stays above the pier. After the concrete has cured, the post drops into that pocket, sits on the steel seat, and is bolted there. Wood never touches concrete or soil. That is the point of the bracket: a buried post rots.

The product that matches this pour is **Sturdi-Wall Plus**, the wet-set bracket made by Perma-Column, in the size for a 3-ply 2×8. A Perma-Column precast column is a different part: a concrete post you set in the hole, with the pocket already on top. Do not try to drop one of those down the 18 in tube. This page is the wet-set bracket.

1. When the pier is poured and the top is at the pad surface, set the bracket on the string line and push the rebar legs down into the wet concrete. Tap it lightly so the concrete closes around the legs. Do not leave air pockets under the steel.
2. Brace the pocket so it cannot tilt or drift while the concrete sets. The post line is the pocket, not the tube. A bracket that leans gives you a leaning post.
3. Perma-Column's wet-set guide says wait 7 days before bolting the wood on, and 30 days before the bracket is at its full design load. The guide is their installation sheet.
4. Cut the post end flat and square. Seat it firmly on the bottom of the pocket. Do not shim it up off the seat. The 2×8 model takes eight ¼ in × 3 in screws, driven without a pilot hole, and two ½ in bolts through the post, tightened to 110 to 120 ft-lb. Drill the bolt hole from both sides of the U if you have to. Do not oval the hole to make the bolt fit.

The bars

These two drawings place the #4 the materials list already counts, about 15 ft per pier. They are a placing helper so the steel is in the hole before the truck comes. They are not a stamped cage. If the engineer adds bars, buy more #4. Do not leave the steel out.

One building pier, #4 bar, tied with wire

Piece	Qty	Cut length	Where it goes
#4 with a 12 in hook	2	5 ft 6 in each	The hook lies in the pad, 3 in above the soil. The leg rises inside the tube and stops 3 in below the concrete top.
#4 straight	2	24 in each	Cross in the bottom of the pad, on top of the hooks, 3 in clear of the pad edge. Tie them to the hooks.
Total	4	15 ft	Two hooked bars are 11 ft. Two straight bars are 4 ft. That is the materials allowance for one pier.

Figure 3. Section through the form, bars in place, before the pour. The tube stops at the top of the pad. The pad under it is solid. The hooks lie in that pad. The legs rise inside the tube. The middle stays open for the bracket's rebar legs.

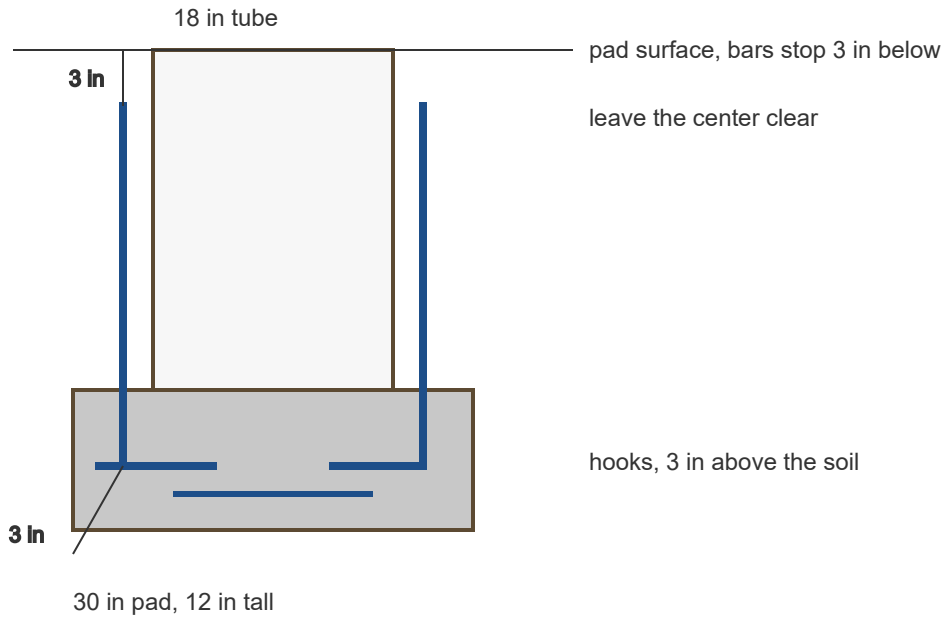
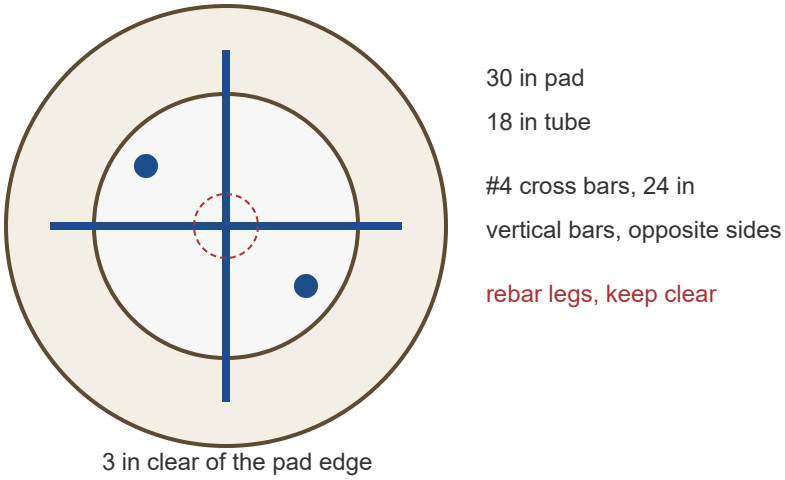


Figure 4. Looking down at the pad. The inner circle is the open tube above the pad, not a hole in the pad. The pad under that circle is solid. The two straight bars cross, 3 in clear of the edge. The dots are the vertical bars. The open middle is for the

bracket's rebar legs. If a leg hits a dot, slide that bar toward the tube wall until it clears, and keep 2 in of concrete between the bar and the tube.



Hang the tied cage from the top of the tube with wire so the hooks stay 3 in off the soil. Do not prop them on a loose stone. Cover against the soil is 3 in. Cover to the tube is at least 2 in. These are ordinary footing covers, not a project-specific design.

The four deck piers

The landing deck has four more holes, drawn as 18 in tubes, 60 in below grade, with post bases rather than the column brackets. The concrete estimate uses the same pad size for them. Dig them in this same pass only if the engineer is keeping that size. Do not set a column bracket in a deck pier, and do not set a deck base in a building pier.

What not to do

- Do not stop at 4 ft because that is as far as the auger goes. Four feet clears the county's 30 in frost depth from the pad surface, but this pier is drawn at 60 in so the pad bears in native soil under the fill. At the north edge, 4 ft from the pad is only about 2 ft into that native soil.
- Do not stand the 18 in tube on the dirt inside the 30 in ring. That pours a doughnut. The column sits on dirt, and the paper keeps the ring of concrete off the column.
- Do not leave the 30 in pad off. The bottom of an 18 in tube has about one-third the bearing area of the 30 in pad.
- Do not pour a pad at the top of an auger hole and call it the footing. The wide part has to be at the bottom.
- Do not add water to the truck mix to make it flow. If the mix is too stiff to fill the ring, tell the driver. Extra water weakens the pier.

Sources

- Detail 3 on the sections page, and the workshop model: 18 in tube, 30 in pad, 12 in thick, bottom 60 in below the pad surface, 4,000 psi, wet-set bracket, 19 building piers. The county frost depth on that detail is 30 in. The 60 in depth is the frost depth plus 30 in below it.
- The pier yardage and the 15 ft of #4 per pier are the materials estimate. Figures 3 and 4 place that 15 ft. They are a helper, not a stamped cage.
- Perma-Column, Sturdi-Wall Plus wet-set installation sheet: the rebar legs go in the wet concrete, the wood waits 7 days, and the 2×8 pocket takes eight screws and two ½ in bolts at 110 to 120 ft-lb. Read November 2026.